THE DIAGONAL OF THE QUOT FUNCTOR IS A CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 729-734 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 736-752, the complete author proof (15 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 19 of queue_moduli.json (pb=802, stmt 729-734, proof 736-752).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 082T: sources/stacks-moduli/pinned-original-source.tex, source lines 752–752. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 083M: sources/stacks-moduli/pinned-original-source.tex, source lines 752–752. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The diagonal of* $\mathrm{Quot}_{\mathcal{F}/X/B} \to B$ *is a closed immersion. If* $\mathcal{F}$ *is of finite type, then the diagonal is a closed immersion of finite presentation.*

Proof. Suppose we have a scheme T/B and two quotients $\mathcal{F}_T \to \mathcal{Q}_i$, $i = 1, 2$ corresponding to T-valued points of $\mathrm{Quot}_{\mathcal{F}/X/B}$ over B. Denote $\mathcal{K}_1$ the kernel of the first one and set $u : \mathcal{K}_1 \to \mathcal{Q}_2$ the composition. By Flatness on Spaces, Lemma 083M there is a closed subspace of T such that $T' \to T$ factors through it if and only if the pullback $u_{T'}$ is zero. This proves the diagonal is a closed immersion. Moreover, if $\mathcal{F}$ is of finite type, then $\mathcal{K}_1$ is of finite type (Modules on Sites, Lemma 082T) and we see that the diagonal is of finite presentation by the same lemma. □

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.